

ArabJobs: A Multinational Corpus of Arabic Job Ads

Mo El-Haj

VinUniversity, Vietnam
Lancaster University, UK
elhaj.m@vinuni.edu.vn
m.el-haj@lancaster.ac.uk

Abstract

ArabJobs is a publicly available corpus of Arabic job advertisements collected from Egypt, Jordan, Saudi Arabia, and the United Arab Emirates. Comprising over 8,500 postings and more than 550,000 words, the dataset captures linguistic, regional, and socio-economic variation in the Arab labour market. We present analyses of gender representation and occupational structure, and highlight dialectal variation across ads, which offers opportunities for future research. We also demonstrate applications such as salary estimation and job category normalisation using large language models, alongside benchmark tasks for gender bias detection and profession classification. The findings show the utility of ArabJobs for fairness-aware Arabic NLP and labour market research. The dataset is publicly available on GitHub: <https://github.com/drelhaj/ArabJobs>.

1 Introduction

The expansion of Arabic Natural Language Processing (NLP) research has supported progress in areas such as sentiment analysis, named entity recognition, and machine translation (Antoun et al., 2020). However, the field continues to face a shortage of datasets that are both linguistically diverse and representative of socio-economic realities. Job advertisements offer a valuable lens into labour market discourse, often encoding assumptions about gender roles, social hierarchies, and regional language practices. Prior research has demonstrated the presence of gender bias in such texts and stressed the importance of computational techniques to detect and reduce these biases (Dikshit et al., 2024a).

Despite the importance of employment-related text for sociolinguistic and fairness-oriented NLP, no publicly available Arabic corpus exists that captures the structure and linguistic diversity of job

advertisements across multiple Arab countries. To our knowledge, no prior datasets have been released in this domain, and existing work on Arabic job-related text is either non-existent or inaccessible. To address this gap, we present **ArabJobs**, a corpus of Arabic job advertisements collected from four countries—Egypt, Jordan, the United Arab Emirates, and Saudi Arabia. The dataset includes structured fields such as job title, location, and salary, as well as unstructured job descriptions, offering broad coverage across sectors and dialects.

2 Literature Review

Despite recent advances in Arabic NLP, the field continues to face a shortage of domain-specific and socio-linguistically diverse corpora. While general-purpose datasets and language models have been developed for Arabic (Antoun et al., 2020; El-Haj, 2020; Alhafni et al., 2024; Daoud et al., 2025; El-Haj and Ezzini, 2024; Elmadani et al., 2025; El-Haj et al., 2024), resources grounded in real-world contexts—such as employment, health, or finance—remain rare. This limits the development of systems capable of modelling Arabic in ways that reflect regional variation, social practices, and occupational language.

For English, job advertisement datasets have enabled a range of impactful studies, particularly in the analysis of bias, fairness, and labour market discourse. For example, recruitment corpora have been used to reveal implicit gender stereotypes in job descriptions (Dikshit et al., 2024b), providing empirical foundations for bias detection tools and fairness-aware text generation. Such work has underscored the value of job ads as a lens into both linguistic and socio-economic structures. However, no comparable resource exists for Arabic, leaving a significant gap in our ability to conduct similar analyses across the Arab region. The ArabJobs corpus fills this gap by introducing the first

publicly available, multi-country corpus of Arabic job advertisements. Covering posts from Egypt, Jordan, Saudi Arabia, and the UAE, it enables the study of regional dialect use, gender representation, and occupational framing in real-world labour discourse. The corpus is designed to support downstream NLP tasks and facilitate investigations into sociolinguistic variation in a structured, professionally relevant setting.

Prior work on gender and dialect in Arabic NLP further highlights the importance of such domain-grounded corpora (Alhafni et al., 2022). Bias detection and mitigation strategies have largely been confined to general-purpose or translated datasets, with limited exploration of high-stakes, real-world domains like employment. Tools such as AraWEAT (Lauscher et al., 2020) and the Arabic Parallel Gender Corpus (Alhafni et al., 2022) provide important foundations for modelling gender sensitivity, while dialect classification benchmarks like MADAR (Bouamor et al., 2019), NADI (Abdul-Mageed et al., 2020), and ALDi (Keleg et al., 2023) offer frameworks for analysing linguistic variation. Yet, these efforts often operate independently of professional or institutional contexts. By anchoring linguistic analysis in the domain of job advertising—where language directly impacts access to opportunity—the ArabJobs corpus offers a new lens for examining structural inequality, dialectal salience, and cultural norms embedded in Arabic textual data. Our study explores how gendered language and job category structures manifest in Arabic job advertisements. We also extend research directions commonly pursued in English NLP, such as implicit gender bias detection and the use of LLMs for salary estimation and job classification—demonstrating how a domain-specific corpus can support analogous investigations in Arabic and open new avenues for NLP research in the region.

3 ArabJobs Corpus

The ArabJobs corpus is the first large-scale, publicly available dataset of Arabic job advertisements, supporting research in NLP, labour market analysis, sociolinguistics, and computational social science. It contains **8,546** ads totalling over **550,000 words**, collected from Egypt, Jordan, Saudi Arabia, and the UAE. These cover a wide range of sectors and reflect regional linguistic and socio-economic variation.

Each entry includes structured fields such as job title, location, salary (or estimate), gender preference, and free-text descriptions. Table 1 presents a breakdown by country, showing the number of ads, gender targeting (male, female, or neutral), and average word count per post. This dataset enables nuanced analyses of how job markets communicate expectations and supports investigations into gendered language, occupational framing, and fairness in employment discourse.

Country	Ads	Male	Female	Neutral	Avg. Word Count
Egypt	3,598	2,085	313	1,200	58.88
Jordan	1,147	498	370	279	47.49
Saudi Arabia	1,854	972	264	618	116.65
UAE	1,947	1,212	427	308	28.57

Table 1: Job Advertisement Statistics by Country

As shown in Figure 1, Egypt and the UAE account for the largest number of job advertisements in the corpus, followed by Saudi Arabia and Jordan. These differences likely reflect underlying labour market dynamics and platform usage across the region. The breakdown also reveals notable variation in posting volume and length, both of which are relevant for downstream analyses of language use and content structure.

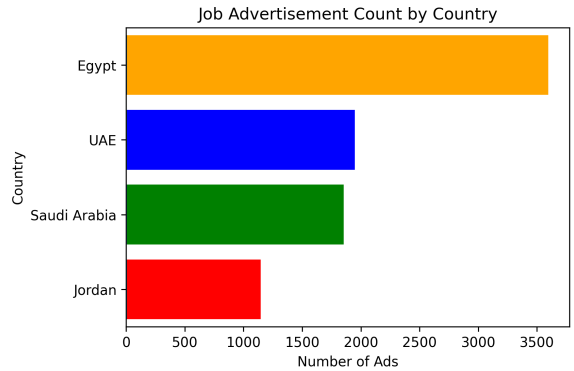


Figure 1: Distribution of job advertisements across four countries in the ArabJobs corpus.

3.1 Data Collection

The **ArabJobs** corpus was constructed by scraping Arabic job advertisements from seven publicly accessible recruitment platforms across the MENA region. We complied with all robots.txt restrictions, excluded paywalled or login-protected content, and implemented rate limiting to ensure respectful data collection. All personally identifiable information—such as names, emails, and phone numbers—was removed during post-processing.

(see Section 8 for further details).

Each job entry in the corpus includes structured fields such as `job_title`, `location`, `salary`, `gender`, `description`, and `country`. Additionally, the dataset contains fields generated via LLM-based inference—`profession`, `salary_local`, `salary_usd`, `job_category`, and `sub_category`—which were subsequently verified by native Arabic-speaking annotators.

4 Dialectal Variation and Code-Switching Analysis

Although ArabJobs does not explicitly annotate dialects, its multinational scope naturally captures regional linguistic variation. To explore this, we conducted an unsupervised analysis using job descriptions from Egypt, Jordan, Saudi Arabia, and the UAE.

We sampled 1,500 ads per country to ensure a balanced dataset and represented job descriptions using TF-IDF features. Dimensionality reduction via Truncated Singular Value Decomposition (SVD) revealed clear regional clusters (Figure 2). Saudi and Emirati ads (Gulf dialects) clustered closely, while Egyptian and Jordanian postings formed separate regions, reflecting variation in dialect and register. For instance, Jordanian ads for female beauty salons often use *صالون سيدات*, whereas terms preferred in Gulf ads include *صالون نسائي* and *صالون حريمي*. Dialectal differences also appear in barbering roles (*مصفف شعر*, *حلاق*, and *كوافير*), as well as in transport-related terms such as *درافتر*, *عجلة*, *دراجة هوائية*, *سكوتر*, *ليسن*, and *رخصة قيادة*.

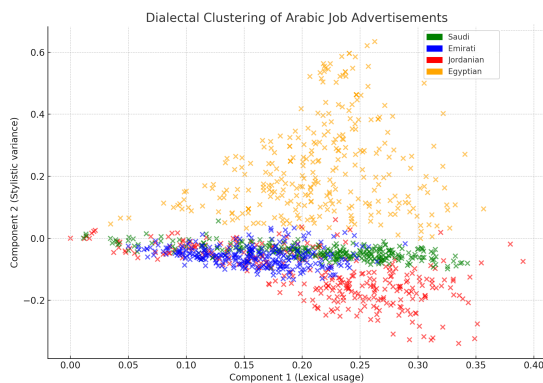


Figure 2: Dialectal clustering. Component 1 captures lexical variation; Component 2 reflects stylistic differences.

We also analysed code-switching—English word usage within Arabic descriptions. As shown

in Figure 3, ads from Jordan, Egypt, and Saudi Arabia featured more English terms (e.g., “Sales Executive”, “Supervisor”), especially in sales and admin roles. In contrast, UAE postings more consistently used Arabic or Arabised terms such as *‘السيرة الذاتية’*, *‘السي في’*, *‘البريد الإلكتروني’*, and *‘إيميل’*.

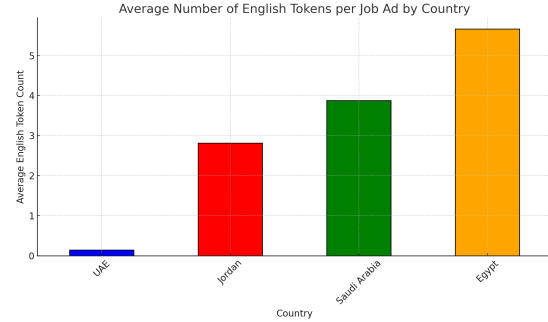


Figure 3: ArabJobs: Arabic-English Code Switching.

While job advertisements are typically composed in Modern Standard Arabic (MSA), dialectal features frequently appear, often unintentionally, even in contexts considered formal. This reflects the broader phenomenon of diglossia in Arabic, where speakers regularly shift between MSA and regional varieties. For instance, Egyptian ads may include everyday colloquialisms such as *عجلة* (“bike”), while Jordanian postings might favour Arabised English borrowings like *سكوتر* (“scooter”). These variations do not necessarily index prestige or class, but rather highlight the influence of local linguistic norms and digital writing practices. Similarly, ads for the beauty sector in Jordan may adopt familiar, community-oriented phrasing, whereas Gulf postings lean towards more formal or gender-specific expressions. These linguistic patterns reflect how Arabic speakers naturally draw from their dialects—even in written form—using them to convey relatability, express culturally grounded meanings, or enhance the communicative effectiveness of the advertisement.

5 Corpus Processing and Normalisation

To enable structured analysis and downstream NLP tasks, we applied several post-processing steps to enrich the raw job advertisements with additional metadata. This included inferring missing salary information, normalising inconsistent job categories, and generating standardised labels. These steps combined rule-based procedures, large language model prompting, and manual verifica-

tion to improve the corpus’s analytical utility.

5.1 Salary Estimation

The salary field records the original salary information when provided, either as a single figure (e.g., 3000 Emirati Dirham) or a range (e.g., 9000–11100 Egyptian Pound). However, only 3,265 job advertisements included this information. To address the substantial number of missing values, we used GPT-4 (OpenAI, 2023) to estimate salaries based on other job attributes. Rather than using the model interactively via a conversational interface, we adopted a prompt-based inference approach. Specifically, we constructed structured prompts that included 3 in-context examples drawn from the 3,265 salary-annotated ads, followed by a new instance requiring prediction (See Appendix A). Examples were reused across prompts, sampled by country and job category to maintain relevance. Our aim was not to introduce a novel estimation method, but to show that the dataset is structured and unambiguous enough to support downstream tasks with state-of-the-art LLMs.

To evaluate the model’s predictive performance, we tested its output against the full set of 3,265 job ads with known salary values. As shown in Table 2, the model achieved a low mean absolute error (MAE) of 11.83 and a root mean square error (RMSE) of 14.84. Additionally, 98.5% of predictions fell within $\pm 10\%$ of the true salary, and 99.45% fell within $\pm 20\%$. The Pearson correlation coefficient was 0.997, indicating a linear alignment in this simulated setup. These results demonstrate that the model performs reliably in structured inference tasks, with prediction quality that aligns well with the distribution of true values.

Metric	Value
Number of Samples Evaluated	3,265
Mean Absolute Error (MAE)	11.83
Root Mean Square Error (RMSE)	14.84
Pearson Correlation (r)	0.997
Within $\pm 10\%$ of Actual Salary	98.50%
Within $\pm 20\%$ of Actual Salary	99.45%

Table 2: Evaluation results for simulated salary estimation using GPT-4

To further validate the reliability of these estimates, we conducted a human evaluation. Two native Arabic-speaking annotators (Annotator 1 and Annotator 2), both fluent in Modern Standard Arabic—independently estimated salaries for a ran-

dom sample of 500 job ads each. Both annotators had access to the full set of 3,265 salary-annotated ads, excluding the 500 samples they were asked to label. As with the model evaluation, salary ranges (e.g., 1000–2000) were reduced to their midpoints for comparison.

Inter-annotator agreement was high: 93% of estimates matched within a $\pm 20\%$ margin, and 89% within $\pm 10\%$. GPT-4’s predictions also aligned well with human judgement. Agreement between GPT-4 and Annotator 1 reached 85% within $\pm 20\%$ and 81% within $\pm 10\%$, while alignment with Annotator 2 was slightly lower at 81% and 78%, respectively. These results, shown in Table 3, demonstrate that the model’s estimates are both stable and broadly comparable in quality to human annotation.

Comparison	Agreement
A1 vs A2 @ $\pm 10\%$	0.89
A1 vs A2 @ $\pm 20\%$	0.93
GPT-4 vs A1 @ $\pm 10\%$	0.81
GPT-4 vs A1 @ $\pm 20\%$	0.85
GPT-4 vs A2 @ $\pm 10\%$	0.78
GPT-4 vs A2 @ $\pm 20\%$	0.81

Table 3: Inter-annotator agreement for salary estimation (A1, A2: Annotators 1 and 2.)

The salary_local and salary_usd columns were generated for all 8,546 job advertisements as explained above. salary_local reflects the salary in the original currency of the job post (e.g., Jordanian Dinar, Saudi Riyal, Emirati Dirham, Egyptian Pound), while salary_usd provides the corresponding amount converted to US Dollars.¹

5.2 Job Category Unification

The job_category field captures the functional sector of each job advertisement (e.g., Customer Service, Engineering). These labels were originally assigned by the source platforms² but varied significantly across sites due to inconsistent taxonomies—for example, موظف استقبال (Receptionist), مساعد إداري (Administrative Assistant), and سكرتير (Secretary) all describe similar roles but were labelled differently. First, all raw category names were aggregated to capture the full range of sectoral variation. Then, GPT-4 was used

¹Conversion rates used: 1 JOD = 1.41 USD, 1 SAR = 0.27 USD, 1 AED = 0.27 USD, 1 EGP = 0.032 USD.

²We preserved the original categorisation in the profession field, as shown in Section 3.1.

to cluster semantically similar labels—translating when needed. For example, خدمة العملاء, and Customer Service / Call Centre were merged under خدمة عملاء (Customer Service).

To reduce fragmentation, rare or overlapping categories were merged under broader labels. For example, طب, تمريض, صيدلة, and علوم ورعاية صحية were unified under الرعاية الصحية (Healthcare). To retain granularity, the original profession labels were preserved in a separate sub_category column, enabling both general and detailed analyses (e.g., comparing nurses and pharmacists).

This process yielded a coherent taxonomy of Arabic job sectors. Table 4 summarises the resulting category distribution.

Arabic Category	English Translation	Ad Count
مبيعات	Sales	1783
فنيين وحرفيين	Technicians and Craftsmen	960
إدارة وسكرتارية	Admin and Secretarial	777
سياحة ومطاعم	Tourism and Restaurants	733
مالية ومحاسبة	Finance and Accounting	579
سيارات وميكانيك	Automotive and Mechanics	460
تسويق	Marketing	447
خدمة عملاء	Customer Service	428
هندسة	Engineering	360
خدمات تنظيف	Cleaning Services	290
موارد بشرية	Human Resources	272
رعاية صحية	Healthcare	260
صناعة وتجارة	Manufacturing and Retail	251
صحة وجمال	Health and Beauty	221
إعلام وتصميم	Media and Design	220
أمن وحراسة	Security	145
سائقين وتوصيل	Drivers and Delivery	145
تعليم	Education	108
تكنولوجيا المعلومات	Information Technology	69
قانون ومحاماة	Law and Legal Services	38
Total	–	8,546

Table 4: Distribution of job advertisements by unified job category

6 Gender Representation and Occupational Trends

The frequent use of gendered language in the ArabJobs corpus makes gender representation and bias a central focus of analysis. Gender is often explicitly stated—e.g., مطلوب موظفة³—or implied through gendered job titles and descriptions. This enables a detailed analysis of both explicit and implicit gender preferences across countries and job sectors.

³In Arabic, grammatical gender is marked morphologically. For instance, موظف (male employee) becomes موظفة (female employee) with the suffix ـة.

It is important to note that gender labels in the dataset are drawn directly from the original job platforms (see Table 1). Our use of the term “implicit gender” does not refer to inferred labels, but rather to gendered language that appears in job descriptions, such as جميلة (“beautiful”) or لبقة (“well-mannered”). By contrast, “explicit gender” refers to ads that state a gender requirement directly, such as through the use of morphologically marked job titles or phrases like مطلوب موظفة (“female employee required”).

6.1 Gender Label Distribution Across Countries

As shown in Figure 4, most job postings are directed at men, with far fewer targeting women or using neutral language. While this imbalance is consistent across countries, its extent varies, reflecting national labour market dynamics and cultural norms, highlighting the need to examine how gender is encoded in recruitment language.

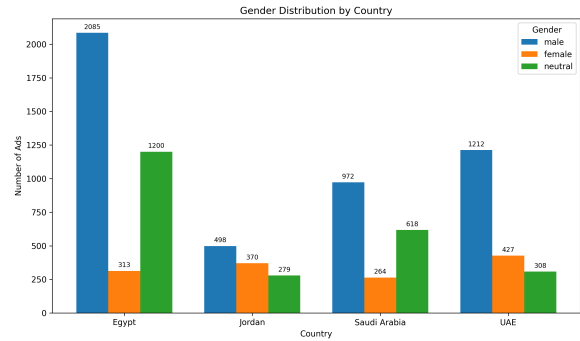


Figure 4: Gender distribution in Arabic job advertisements by country.

6.2 Gendered Occupational Patterns

The corpus spans a wide range of occupational diversity making it suitable for downstream NLP tasks involving profession classification, summarisation, and thematic bias detection. A closer analysis, however, reveals clear gender-based occupational segregation.

As shown in Figure 5, male-targeted job ads disproportionately reference technical, physical, and logistical professions—such as فنيين (technicians), مهندسين (engineers), سائقين (drivers), and مندوب مبيعات (sales agents). Industry-related roles such as أعمال ميكانيكا (mechanical work), إنتاج (manufacturing), and مقاولات (construction) are also dominant. These roles tend to prioritise skills related to physical labour, trade certifications, and lo-

[illegible]

In contrast, the female-targeted word clouds in Figure 6 reveal a concentration in service, administrative, and care-related roles. Commonly mentioned positions include سكرتارية (secretarial work), تجميل (beauty), خدمة العملاء (customer service), مساعد إداري (administrative assistant), and موظفة استقبال (receptionist). These roles typically emphasise communication, hospitality, appearance, and interpersonal skills—reinforcing prevailing gender norms in the professional landscape.



6.3 Gender-Based Salary Disparity

To quantify gender-based pay disparities, we compute the gender pay gap as the difference between the average salaries of male- and female-targeted job advertisements, relative to the female average:

Country	Gender	AvgLoc	AvgUSD	N
Egypt	female	7079.29	226.54	313
Egypt	male	8080.22	258.57	2085
Egypt	neutral	8078.95	258.53	1200
Jordan	female	358.92	505.98	370
Jordan	male	412.73	581.95	498
Jordan	neutral	403.48	568.92	279
Saudi Arabia	female	4057.12	1095.43	264
Saudi Arabia	male	4356.65	1176.3	972
Saudi Arabia	neutral	4060.97	1096.47	618
UAE	female	3092.28	834.96	427
UAE	male	2641.01	713.08	1212
UAE	neutral	2998.43	809.61	308

A positive gap indicates that men are offered higher average salaries than women, while a negative value signals the reverse. As shown in Table 6, male-targeted roles have higher average pay in Egypt (14.14%), Jordan (15.01%), and Saudi Arabia (7.38%). The UAE is the exception, showing a negative gap of -14.6%, where female-targeted roles offer slightly higher salaries. This is largely due to sectoral distribution: the most common category in UAE ads is فنيين وحرفيين (Technicians and Craftsmen), comprising 18% of all postings and offering the lowest average pay—mostly targeted at men. .

Country	M-USD	F-USD	Gap\$	Gap%
Egypt	258.57	226.54	32.03	14.14%
Jordan	581.95	505.98	75.97	15.01%
Saudi Arabia	1176.3	1095.43	80.88	7.38%
UAE	713.08	834.96	-121.88	-14.6%

6.4 Structural Gender Representation Across Job Categories

³These figures reflect unregulated online job postings and may not represent official labour market policies.

tween male and female shares—to capture the degree of gender exclusivity.

Table 7 presents the results, ranked by descending gender skew. Certain fields show extreme male dominance, such as أمن وحراسة (Security) and فنيين وحرفيين (Technicians and Craftsmen), with over 96% of postings targeting men. Others, like صناعة وتجارة (Manufacturing and Retail) and سائقين وتوصيل (Drivers and Delivery), also display substantial male bias.

In contrast, categories like صحة وجمال (Health and Beauty) and تعليم (Education) are predominantly female-oriented, with over 70% of postings directed at women. These patterns reflect deeply embedded gender norms around occupational roles.

The analysis shows that gender disparity is not limited to salaries—it is structurally rooted in the allocation of roles. Addressing gender equity in the labour market requires tackling both pay gaps and access to opportunity.

Arabic Category	English	All	%Male	%Female	Dominance	Skew (%)
سيارات وميكانيك	Automotive and Mechanics	425	98.4	1.6	Male	96.8
أمن وحراسة	Security	118	98.3	1.7	Male	96.6
سائقين وتوصيل	Drivers and Delivery	124	97.6	2.4	Male	95.2
فنيين وحرفيين	Technicians and Craftsmen	869	96.5	3.5	Male	93.0
هندسة	Engineering	309	91.3	8.7	Male	82.6
موارد بشرية	Human Resources	184	89.1	10.9	Male	78.2
صناعة وتجارة	Manufacturing and Retail	196	88.3	11.7	Male	76.6
مالية وحاسبة	Finance and Accounting	306	81.4	18.6	Male	62.8
سياحة ومطاعم	Tourism and Restaurants	499	79.8	20.2	Male	59.6
تكنولوجيا المعلومات	Information Technology	34	79.4	20.6	Male	58.8
تعليم	Education	66	22.7	77.3	Female	54.6
خدمات نظيف	Cleaning Services	231	74.9	25.1	Male	49.8
مبيعات	Sales	1082	74.5	25.5	Male	49.0
إعلام وتصميم	Media and Design	93	74.2	25.8	Male	48.4
تسويق	Marketing	304	74.0	26.0	Male	48.0
رعاية صحية	Healthcare	192	66.1	33.9	Male	32.2
صحة وجمال	Health and Beauty	206	36.4	63.6	Female	27.2
خدمة عملاء	Customer Service	297	37.0	63.0	Female	26.0
إدارة وسكرتارية	Admin and Secretarial	587	62.9	37.1	Male	25.8
قانون ومحاماة	Law and Legal Services	19	57.9	42.1	Male	15.8

Table 7: Gender skew across job categories, measured as the absolute difference between male and female ad proportions.

To better understand salary distribution across job categories, we visualised the average salaries for male- and female-targeted job advertisements, paying particular attention to dominant gender representation. Many professions show strong gender imbalances—for example, 98% of أمن وحراسة ads target men—so simply averaging all ads could produce misleading results. To account for this, we applied a dominance-aware adjustment strategy.

We began by computing the average salaries separately for male-targeted and female-targeted ads within each category. For each category, we identified the dominant gender based on the number of advertisements. The dominant gender’s average salary was then given greater interpretive weight to minimise distortion from underrepresented groups. Figure 7 illustrates this compari-

son. The salary lines for men (solid) and women (dashed) vary across categories, with the grey bars showing the adjusted category-wise averages weighted by gender dominance.

The analysis reveals that high-paying fields like Engineering (هندسة) and IT (تكنولوجيا المعلومات) are predominantly male-targeted, with female ads in these sectors offering considerably lower average salaries—though such cases are few. In contrast, Education (تعليم), typically female-dominated, shows higher average pay for women, likely due to a small number of well-paid positions. Sales (مبيعات) and Customer Service (خدمة عملاء) are more gender-balanced and exhibit narrower salary gaps. Security (أمن وحراسة) and Drivers and Delivery (سائقين وتوصيل) remain male-exclusive, rendering female salary data in these fields negligible. Interestingly, sectors like Marketing (تسويق) and Health and Beauty (صحة وجمال) offer higher average pay for female-targeted roles, though male participation in these fields is limited. Overall, gender disparities persist not only in pay but also in access to lucrative professions, with many seemingly positive trends for women arising from isolated cases rather than systemic equality.

7 Linguistic Bias in Arabic Job Ads

To better understand the linguistic framing surrounding gender-targeted language in Arabic job advertisements, we conducted a concordance analysis using a window of ± 4 words around selected gendered or appearance-related terms. The analysis was based on tokenisation using the CAMEL Tools Arabic tokenizer for improved segmentation quality (Obeid et al., 2020). Our analysis of Arabic job advertisements reveals a concerning pattern of linguistic bias, especially in job posts targeting women. A range of ads explicitly require candidates to meet criteria unrelated to professional qualifications or experience, focusing instead on appearance, age, and marital status. Table 8 summarises the most frequent patterns we observed.

These include phrases such as جميلة، حسنة المظهر، عزباء، شابة، لبقّة، أنيقة and expressions that specify age limits (e.g., 22 و 30 العمر بين) or prefer candidates who are متفرغة (i.e., fully available), sometimes adding that they must be not married (غير متزوجة) (single).

Such language reinforces stereotypes about physical attractiveness and gender roles, particularly in roles such as receptionist, sales assistant,

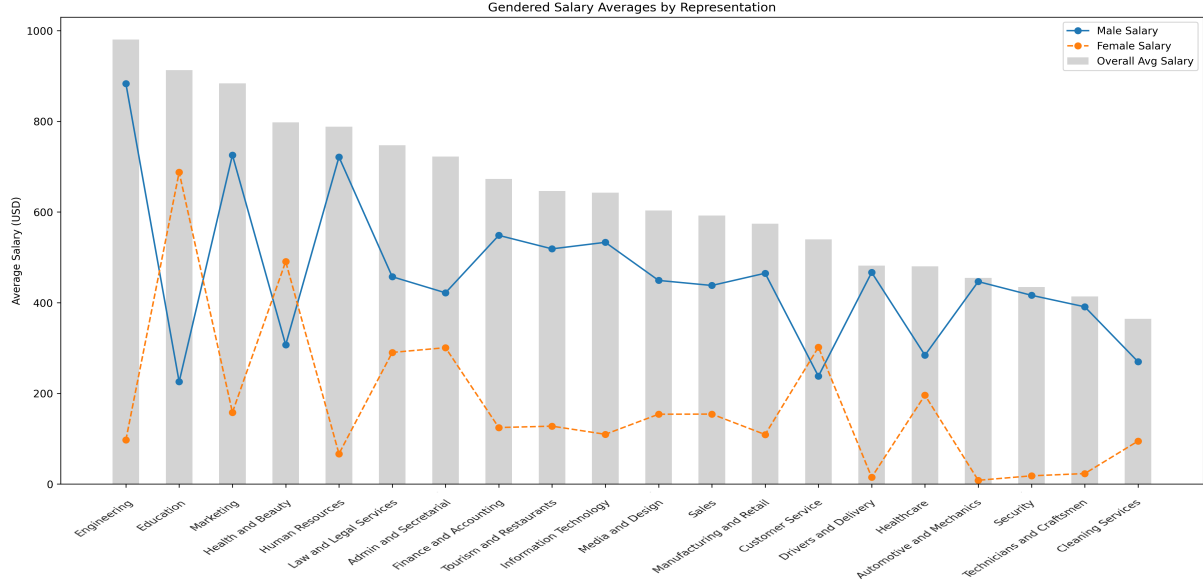


Figure 7: Average salaries in USD by job category, separated by gender and normalised for representation. Bars show overall average; lines indicate male and female-specific averages.

Bias Type	Examples from Ads
Appearance	جميلة، أنيقة، حسنة المظهر، مظهر لائق، مرتبة، غير محجبة beautiful, elegant, good-looking, decent appearance, tidy, not veiled
Personality	لبقة، لبق، لينة في التعامل، شخصية قيادية، لينة بالصدت polite, articulate, good at interaction, leadership personality, well-spoken
Age Limits	العمر لا يتجاوز 52 سنة، من 18 إلى 30 سنة، العمر بين 22 و 35 age must not exceed 52, from 18 to 30, age between 22 and 35
Marital/Availability Status	عزباء، متفرغة للعمل، غير متزوجة single, available for work, not married
Gender Filters	ذكور فقط، إناث فقط، مطلوب شاب، يفضل شابة males only, females only, young man wanted, young woman preferred
Emotion/Soft Skill Framing	لبقة، حسنة السلوك، وجه يشوش، حنون، لينة مع الزبائن articulate, well-behaved, cheerful face, kind-hearted, good with customers

Table 8: Examples of Biased Criteria in Arabic Job Advertisements

or spa worker. Furthermore, certain phrases demand emotional traits like being *لبقة* (polite/eloquent), which often surface alongside gendered expectations. These requirements, especially when associated with low-skilled roles, suggest systemic patterns of bias and discrimination in hiring.

These phrases indicate structured and recurring forms of discrimination in employment language. A larger sample of concordance examples is included in Appendix B to support transparency and enable further qualitative inspection.

8 Conclusion

This paper introduced **ArabJobs**, the first large-scale, publicly available corpus of Arabic job advertisements spanning four Arab countries. The dataset captures linguistic, regional, and socio-economic variation across over 8,500 postings and

provides a valuable resource for studying gender representation, dialectal diversity, and occupational language in Arabic. The findings not only validate the quality and versatility of the corpus but also highlight its broader potential to support fairness-aware NLP in under-resourced, real-world contexts. Through a series of experiments, we demonstrated the utility of the dataset for downstream tasks such as salary estimation, job classification, and bias detection. Our analyses revealed systematic gender disparities in both language use and pay, along with clear patterns of occupational segregation. We further showed that large language models like GPT-4 can reliably estimate missing salary information and produce predictions closely aligned with human judgement, reinforcing the value of LLMs in socio-economic text analysis and structured inference.

Ethical Considerations

The ArabJobs corpus was collected from publicly accessible websites that did not require authentication or payment. Although available in the public domain in practice, the listings are not covered by formal open data licences (e.g., Creative Commons), so the corpus is distributed under a research-only licence for non-commercial academic use. We do not claim ownership of the original content.

All scraping was conducted in compliance with the `robots.txt` directives of the source sites, and

no automated access was made to restricted paths. Personally identifiable information was stripped from all records to ensure responsible and ethical data handling.

Table 9 lists the data sources and scraping constraints observed at the time of collection.

Website	Scraping Allowed?	Notes
naukrigulf.com	Yes	Avoid listed disallowed paths
gulftalent.com	Yes	Do not impersonate blocked bots
dubizzle.com	Yes	Avoid disallowed paths, rate-limited
tangeeb.com	Yes	Avoid URLs with parameters
jordanrec.com	Yes	Avoid admin/plugin paths
forasna.com	Yes	Avoid query filters in URLs
sabbar.com	Yes	Fully allowed; provides job sitemaps

Table 9: Scraping permissions and constraints for the ArabJobs corpus sources.

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Appendix A: Example Prompt for Salary Estimation

Below is a simplified illustration of the structured few-shot prompt used with GPT-4. Three examples with known salaries are provided, followed by one target ad requiring prediction.

SYSTEM: You are an assistant that predicts monthly salaries for job ads in Arabic-speaking countries . Always return the salary as <number> <currency>.

EXAMPLE 1
Title: Accountant
Location: Cairo, Egypt
Category: Finance and Accounting
Gender: Any
Description: Responsible for financial reports and invoices.
Salary: 9,500 EGP

EXAMPLE 2
Title: Sales Executive
Location: Riyadh, Saudi Arabia
Category: Sales
Gender: Male
Description: Outdoor sales for electronics company.
Salary: 6,500 SAR

EXAMPLE 3
Title: Nurse
Location: Amman, Jordan
Category: Healthcare
Gender: Female
Description: Provide patient care in hospital setting.
Salary: 720 JOD

PREDICT
Title: HR Assistant
Location: Dubai, UAE
Category: Human Resources
Gender: Any
Description: Support recruitment and employee records.
Salary:

Appendix B: Job Ads Bias Concordances

التفاصيل	الصفة	التفاصيل
مؤهل عالي - احوال ادارية مطلوب سكريرة	حسنة	المظهر للعمل في شركة مقاولات
الإنكليزية، المهام الرئيسية تشمل: • لياقة و	حسنة	المظهر • استقبال وتوجيه الزوار والضيوف •
استقبال في معرض جربة في التعامل	حسنة	المظهر لها خبرة في التحدث
بحر مدبرة مكتب بالشروط التالية:خبرة،	حسنة	المظهر.ليقة.التفرغ.العمل بمؤسسة
شارح المطار مطلوب سكريرة خدمة عملاء	حسنة	المظهر داخل مدينة أوطي قريب
لشركة استثمار في أوطي آل تبيان	حسنة	المظهر متفرغ للعمل تجيد مبرات
المطار مطلوب سكريرة على وجه سرعة	حسنة	مظهر وبشكل انجليزي راتب اقامة
والتنفيذ وإعداد التقارير والبحث عبر الانترنت	حسنة	المظهر وتنفع بدقة العمل والالتزام
دور الساعة تجيد التسويق والتصميم تكون	حسنة	المظهر بحث عن سكريرة جادة
في كوفي شوب داخل المصنف 54 شرط	حسنة	المظهر مطلوب موظفة للعمل وإتير
متزله ينزل في أوطي شارع الدفاع	حسنة	المظهر مطلوبة عاملة منزلية ميت
للعمل بكافة في بحمان راشيديه 1 الراتب 2800	حسنة	المظهر ليقه عدد ساعات العمل 10
قطر)، وفق الشروط التالية: أن تكون	ليقة	في التعامل وذات مظهر لائق.
خبرة في الاستقبال والتعامل مع العملاء	ليقة	بالتعامل بآداب شيري ومكافآت شيرة
الشركات العالمية والتعامل معها بإتخافية. شخصية	ليقة	ومبرات تواصل وتعظم عالية. التوظيف
اجادة استخدام الكمبيوتر وبراع الأوفس	ليقة	في التعامل ومظهر مناسب الدوام
الإنجليزية (قراءة وكتابة ومحادثة) شخصية	ليقة	مظهر أيق واخترافي العمر لا
العمر بين 22- 28 سنة - على أن تكون	ليقة	في التعامل ذو مظهر لائق
على جميع الوثائق في قسم الاستقبال	مرتبة	التقديم خبرة في خدمة العملاء.
منطقه أم علفة أن تكون أيقه	مرتبة	جربة تجيد العمل في مجال
الانكليزي بأن تكون حسنة المظهر أيقه	مرتبة	للعمل في مؤسسة كالا لتصارة
والتعامل باليشرة الرجال ليقه في التعامل وحسنة المظهر	بدون حجاب	8 ساعات عمل 6 أيام عمل ويوم راحة
المهام والتعامل مع الأوراق المشفرة. - ان تكون حسنة المظهر و	غير محجبة	كون بيئة العمل في بيوتي ستر والتعامل مع العملاء.
مبرات تواصل متميزة وهدرة على التنسيق والمتابعة.	بدون حجاب أو حجاب Modern	مكان العمل: شارع الجازدز - عمان الدوام.

Table 10: Biased Criteria in Arabic Job Advertisements

حسنة ، ليقه ، مرتبة، غير محجبة -

التفاصيل	الصفة	التفاصيل
المنطقة اوالمناطق المجاوره إسطنبول ان تكون	عزباء	رقم التواصلقدم على الوظيفة لإظهار
توفر لأب توب العمر بين 20-25 سنة	عزباء	معرفة جيدة في استخدام الكمبيوتر
شابه المدرسة خاصة في الزرقاء ويفضل	عزباء	ومتفرغة للعمل ولديها معرفة باستخدامات
رصيفه من عمر 18-30 الشرط ان تكون	عزباء	للتواصل على الواتس اب مطلوب
في العمل. الشروط المطلوبة: سعودية الجنسية	متفرغة	للعمل القدرة على العمل في
مواييل يفضل من سكان جبل الحسين	متفرغة	ليقة رد آلي واتسقدم على
ساكنة بالقرب من السوق * ان تكون	متفرغة	للعمل غاماً قدم على الوظيفة
السبت للخميس براتب 290 على ان تكون	متفرغة	وسريعة التعلم (طبيعة العمل سكرتاريا
وبراج التواصل - شفت في مجال الأزياء -	متفرغة	للعمل - تجيد التعامل مع العملاء -
المدينة الرياضية وضواحيها 2- ان تكون	متفرغة	وغير متزوجة . 3- ليقه في التعامل

Table 11: Biased Criteria in Arabic Job Advertisements

متفرغة ، عزباء -

التفاصيل	الصفة	التفاصيل
يتوفر فيه الشروط التالية: - أن يتراوح	العمر	بين 24 إلى 28 سنة - أن يمتلك
ان يتوفر الشروط التالية للتقدم للوظيفة *	العمر	من 20 ل 30 * مكان السكن بالقرب
كأين صالة خبرة سنتين على الأقل	العمر	لا يقل عن 25 -موظفين صالة خبرة
موظفين صالة خبرة سنة على الأقل	العمر	لا يقل عن 20 -موظفين بارستا بارد
في العبدلي - خبرة في المجال الطبي -	العمر	أقل من 30 - راتب اول ثلاث
يفضل وجود خبرة باستخدام الكمبيوتر الشروط: -	العمر	من 25 لغايه 35 سنه - أن يكون
بارد وساخن خبرة سنتين على الأقل	العمر	لا يقل عن 22 عام -موظفين أراجيل
MV وواجهات خبرة إدارة والتنسيق والمشتريات	العمر	من 25 سنة فما فوق رواتب
وساعات تمتلك خبرة في مجال المبيعات	العمر	من 25 - 37 ذات مظهر لائق في
يحد أدنى في مجال اعداد المشاريع 3.	العمر	بين 21-35 سنة 4. القدرة على التعامل
يكون متواجدا في بحمان أو الشارقة.	العمر	بين 25 و30 سنة يحمل رخصة
خبرة سابقة في استخدام الرافعه الشوكية	العمر	لا يزيد عن 45 سنة سكان
حلاق ليق في التعامل مع الزبائن	العمر	من 18 - 30 سنة للعمل بالقرب من
قوية واللباقة في التعامل مع العملاء.	العمر	بين 20 سنة، مزاييا العمل: رواتب
مطعم نسائي في العين في الجاهلي	العمر	أقل من 40 ولديها خبرة في
عن سنة في نفس المجال لايزيد	العمر	عن 35 عاما يفضل سكان عمان
صافي + العمولة.متطلبات الوظيفة :- مؤهل مناسب .-	العمر	من 22 إلى 30 سنة- يفضل ان
بالانضباط والقدرة على العمل الشروط العامة: •	العمر	من 18 إلى 30 سنة فقط • الالتزام
المرتب ٠٠٣ صافي + العمولة.متطلبات الوظيفة :مؤهل مناسب •	العمر	من 22 إلى 32 سنةمكان العمل : مدينة
صناعية هيلي يحمل اقامة قابلة للإعارة	العمر	لا يتجاوز 25 شرط سوري الجنسية
المدينة الرياضية وضواحيها . 2- ان يكون	العمر	من 25 - 35 3-أن يمتلك الخبرة و
تنظيف مجال التنظيف المنزلي بالشروط التالية :	العمر	من 30 لغاية 50 الإلتزام و الجدية
تسويق ومبيعات استيراد وتصدير العمل فوري	العمر	مطلوب 47 سنه فقط عداد السيدات
بكالوريوس في أي تخصص ذو صلة.	العمر	ما بين 30 إلى 40 سنة. خبرة
القدرة على إدارة الحسابات والمالية للشركة	العمر	من 23 إلى 35 انضم إلنا كدير
التدريب المهني * اجاده القرائه و الكآله	العمر	حتى 35 على من تنطبق عليه

Table 12: Biased Criteria in Arabic Job Advertisements

العمر -